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ERG Analysis API v2.4.0

Tier 3 — T3-A: ISCEV Compliance Checklist

Document ID	T3-A_ISCEV_Compliance_Checklist_v1_0
Project	ERG Analysis API — AI Fairness CIO (Charity Reg. No. 1218464)
Pipeline version	v2.4.0 (PIPELINE_VERSION = "2.4.0")
Pipeline file	erg_v2_4_0.py (Google Colab notebook; Flask server port 8080)
Report generator	erg_report_generator.py (ReportLab PDF, v2.4.0)
ISCEV standard	McCulloch DL et al. (2015). ISCEV Standard for full-field clinical electroretinography. <i>Doc Ophthalmol</i> 130(1):1–12. DOI: 10.1007/s10633-014-9473-7
ISCEV protocol ref	Robson AG et al. (2022). ISCEV Standard for full-field clinical electroretinography — 2022 update. <i>Doc Ophthalmol</i> 144(3):165–177.
Normative reference	Baker et al. (2025). <i>Doc Ophthalmol</i> 150(2):47–64. DOI: 10.1007/s10633-025-10009-2. N=407.
OSF Pre-registration	https://doi.org/10.17605/OSF.IO/6WA42
Author	Hamid Tavakoli, Founder & CEO, AI Fairness CIO
Date	24 June 2026
Status	Signed-off by Hamid Tavakoli

1. Scope and Purpose

This document records the results of the T3-A ISCEV Compliance Audit for ERG Analysis API v2.4.0. Each checklist item maps one ISCEV 2022/2015 requirement to the corresponding implementation in `erg_v2_4_0.py` and `erg_report_generator.py`, assigns a status (PASS / AMBER / FAIL / PENDING / N/A), and records any required remediation action.

The audit covers five domains:

- Recording parameters (sampling rate, bandwidth, pre-stimulus baseline, post-stimulus duration)
- Signal conditioning (bandpass filter, notch filter policy, spike removal)
- Waveform analysis (protocol-specific measurement windows, a-wave, b-wave, OPs, PhNR, flicker)
- Report and output (parameter labels, units, FHIR coding, disclaimer, traffic-light classification)
- Quality grading and compliance flags (quality grade, ISCEV compliance flag in PDF and FHIR output)

2. Reference Framework

Primary standard: McCulloch DL, Marmor MF, Brigell MG, et al. ISCEV Standard for full-field clinical electroretinography (2015 update). *Doc Ophthalmol* 130(1):1–12. DOI: 10.1007/s10633-014-9473-7.

2022 update: Robson AG, Tufail A, Fitzke FW, et al. ISCEV Standard for full-field clinical electroretinography — 2022 update. *Doc Ophthalmol* 144(3):165–177.

Note on version labelling in `erg_v2_4_0.py`: the pipeline configuration header reads “ISCEV 2022 Compliant” and all inline comments cite “Page 7, Col 2” with paragraph references. These page references are consistent with Robson et al. (2022) and are used throughout this checklist. The locked reference for waveform parameter thresholds and normative Z-scores is Baker et al. (2025).

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3. Findings Summary

Result	Count
PASS	14
AMBER	3
FAIL	1
PENDING	1
N/A	6
TOTAL	25

Key findings:

- All hard ISCEV recording parameter requirements (sampling rate, bandwidth, pre-stimulus, post-stimulus) are implemented correctly and enforced at runtime.
- Notch filter default OFF is correctly implemented and gated on explicit user consent, fully compliant with ISCEV 2022 page 7 paragraph 4.
- Protocol-specific measurement windows for all five standard protocols (DA 0.01, DA 3, DA 10, LA 3, LA 30 Hz) are present and correctly bounded.
- One FAIL is raised: the pipeline version string in the CONFIG header reads “Version 2.3.2” on line 13 (comment block) while PIPELINE_VERSION = “2.4.0” on line 111. The comment must be corrected to avoid version ambiguity in audit logs.
- One AMBER is raised for age-group handling: the ERGConfig.AGE_GROUPS list (line 94–97) includes paediatric strata (0–12mo, 1–5y, etc.) that have no Baker 2025 normative equivalents. The pipeline emits a WARNING flag for these groups, but the PDF report does not surface a visible paediatric disclaimer block. A user-visible warning is recommended.
- One AMBER is raised for the FHIR code system URL: it points to the GitHub repository rather than a versioned, persistent URI. For regulatory submissions, a versioned CodeSystem resource should be registered.
- Layer 3 Specialist output is marked PENDING in both the pipeline and report generator, correctly indicating this is aspirational (Chapters 8/14) and not yet implemented. This is a known pre-registered limitation and is classified as PENDING, not FAIL.

4. Detailed Checklist

Column guide: # = checklist item ID; Status = PASS | AMBER | FAIL | PENDING | N/A. All code references are to `erg_v2_4_0.py` unless otherwise stated.

#	ISCEV Requirement	Pipeline Implementation	Status	Notes / Actions
A — Recording Parameters (ISCEV 2022 Page 7, Col 2)				
A1	Sampling rate ≥ 1000 Hz (Para 2)	ISCEV_MIN_SAMPLING_RATE_HZ = 1000.0 (line 29). Enforced in <code>validate_iscev_requirements()</code> (line 274–302). Runtime flag <code>sampling_rate_ok</code> written to audit JSON.	PASS	Pipeline operates at <code>fs = 2000 Hz</code> , which exceeds the ISCEV minimum of 1000 Hz and is fully compliant. A runtime warning is raised in the Layer 4 audit output only if <code>fs < 1000 Hz</code> is submitted.
A2	High-pass filter ≤ 0.3 Hz (Para 3)	ISCEV_HIGHPASS_HZ = 0.3 (line 27). Applied in <code>ERGFilter.apply_bandpass()</code> (line 595–601) via 4th-order Butterworth zero-phase filter (<code>sosfiltfilt</code>).	PASS	Value locked in <code>ERGConfig</code> and cannot be overridden at runtime without code change.
A3	Low-pass filter ≥ 300 Hz (Para 3)	ISCEV_LOWPASS_HZ = 300.0 (line 28). Applied as upper bound of bandpass. Hardware	PASS	Hardware cutoff detection is conservative (0.95 factor) to prevent aliasing.

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#	ISCEV Requirement	Pipeline Implementation	Status	Notes / Actions
		cutoff detection (line 638–641) reduces effective cutoff if hardware is band-limited.		
A4	Pre-stimulus baseline ≥ 20 ms (Para 5)	ISCEV_MIN_PRESTIMULUS_MS = 20.0 (line 30). Validated in validate_iscev_requirements(). get_prestimulus_status() returns ADEQUATE / PARTIAL / INSUFFICIENT.	PASS	Status propagated to Layer 4 audit and FHIR output. PARTIAL warning displayed if 10–19 ms.
A5	Post-stimulus recording ≥ 300 ms (Para 5)	ISCEV_MIN_RECORDING_MS = 300.0 (line 31). Validated against (total_duration_ms - prestimulus_ms). Flag post_stimulus_ok in audit JSON.	PASS	Generator ERG_CSV_Generator_v2_4_2.py produces 350 ms total (300 ms post-stimulus). T2-E Tier 2 re-validation confirmed 30/30 PASS.
A6	Flash duration < 5 ms (Page 4, Col 2)	ISCEV_FLASH_MAX_DURATION_MS = 5.0 (line 33). Not enforced at runtime (hardware parameter). Flash duration is an input widget (flash_duration_widget) with no upper-bound validation check.	AMBER	ACTION: Add runtime validation: if flash_duration_ms ≥ 5.0 raise a Layer 4 warning flag flash_duration_noncompliant = True.
B — Signal Conditioning (ISCEV 2022 Page 7, Col 2, Para 4)				
B1	Notch filter OFF by default (Para 4)	NOTCH_DEFAULT_STATE = False (line 41). run_filter_pipeline() only applies notch if apply_notch=True AND user_confirmed_notch=True (line 631). Filter log records “Skipped (ISCEV 2022 compliant)” when not applied.	PASS	Double-gate (widget + consent) prevents accidental application. Filter log preserved in Layer 4.
B2	Notch filter warning if applied (Para 4)	When notch is applied, filter log writes “WARNING: Notch filters distort ERG waveform per ISCEV 2022” (line 634).	PASS	Warning appears in Layer 4 technical audit output and in the PDF report filter log section.
B3	OP band: 75–300 Hz (Para 4)	ISCEV_OP_BAND_LOW_HZ = 75.0 (line 32); OP_BAND_HIGH_HZ = 300.0 (line 65). extract_ops() applies bandpass 75–300 Hz (line 667–669). OP SNR assessed against 3 dB threshold.	PASS	Consistent with ISCEV 2022 and Baker 2025 OP methodology.
B4	Spike / artefact removal documented	Median filter (kernel = 5 ms, line 39) applied first in every run_filter_pipeline() call. Spike count, max spike amplitude, and drift flagged in Layer 4 artifact audit (line 470–538).	PASS	Spike removal is always applied. Results logged per recording.
B5	Zero-phase filtering for offline analysis	sosfiltfilt() used throughout (line 601, 613, 669). apply_streaming_filter() raises NotImplementedError (line 660–665) to prevent accidental causal-filter use.	PASS	Streaming pathway correctly blocked in v2.4.0.
C — Waveform Analysis (ISCEV 2022 Protocols)				
C1	DA 0.01 protocol: b-wave only (rod-isolated)	Protocol 'DA 0.01': a_wave_window_ms = (None, None); b_wave_window_ms = (40, 130) ms (line 700–701). a-wave amplitude returned as NaN.	PASS	Consistent with ISCEV 2022: DA 0.01 stimulus does not generate a measurable a-wave.
C2	DA 3 protocol: a-wave and b-wave (mixed response)	Protocol 'DA 3': a_wave_window_ms = (10, 35); b_wave_window_ms = (30, 100) ms (line 703–704). b-wave search starts after a-wave trough (line 748–752).	PASS	b/a ratio computed for DA 3 only; normal range 1.8–3.5 (modelled mean=2.65, SD=0.425).
C3	DA 10 protocol: a-wave and b-wave (high-intensity rod-cone)	Protocol 'DA 10': a_wave_window_ms = (8, 30); b_wave_window_ms = (25, 95) ms (line 705–706).	PASS	Window bounds within published ISCEV typical ranges for DA 10.
C4	LA 3 protocol: photopic a-wave, b-wave, PhNR	Protocol 'LA 3': a_wave_window_ms = (12, 28); b_wave_window_ms = (25, 60) ms (line 707–708). PhNR extraction implemented with polarity check and atypical polarity flag.	PASS	PhNR atypical polarity raises Layer 4 flag and red warning block in PDF report.
C5	LA 30 Hz flicker protocol: b-wave peak only	Protocol 'LA 30 Hz': a_wave_window_ms = (None, None); b_wave_window_ms = (20, 60) ms (line 711–713).	PASS	Consistent with ISCEV 2022 flicker protocol — a-wave not defined for 30 Hz stimulus.

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#	ISCEV Requirement	Pipeline Implementation	Status	Notes / Actions
C6	a-wave amplitude sign convention (trough, negative)	a-wave forced negative via <code>-abs()</code> at end of <code>extract_away_bwave()</code> (line 774–775). Double-negation bug resolved in v2.4.0 (Tier 1 defect A1/S03).	PASS	Validated across all 41 synthetic cases in Tier 1 (41/41 PASS).
C7	b-wave amplitude as trough-to-peak distance (positive)	<code>b_wave_amplitude = b_wave_raw - a_wave_amplitude</code> (line 764). Forced positive via <code>abs()</code> if result is negative (line 766–767).	PASS	Clinically correct: b-wave amplitude is measured from a-trough to b-peak.
C8	OP measurement: OP2, OP3, OP4 with correct windows	OP windows: OP2 15–25 ms, OP3 25–35 ms, OP4 35–65 ms post-stimulus (verified Tier 1). OP sum = OP2+OP3+OP4. Applied to 75–300 Hz bandpass-filtered signal.	PASS	OP4 window extended to 65 ms in v2.4.0 to capture slower potentials.
D — Z-Score Classification and Traffic Light (Baker et al. 2025)				
D1	Normative reference: Baker et al. (2025) N=407, three age strata	REFERENCE_RANGES loaded from <code>data/normative_data_baker2025.json</code> at <code>ERGReportGenerator</code> instantiation (line 1063–1119). Strata: <code>le35</code> ($\leq 35y$), <code>36to59</code> (36–59y), <code>ge60</code> ($\geq 60y$).	PASS	Normative data externalised to JSON (T2-A). <code>FileNotFoundException</code> raised if file absent (line 1089). Reference source string written to Layer 3 <code>normative_meta</code> .
D2	DTL Deming regression transform applied to DTL amplitudes only	<code>_apply_dtl_transform()</code> inverts Baker 2025 Table 2 regression ($GFE = (STE - intercept)/slope$) (line 1154–1191). Applied to <code>a_amp</code> and <code>b_amp</code> only; implicit times NOT transformed.	PASS	Transform flag <code>_dtl_deming_correction_applied</code> written to Z-scores dict for Layer 4 traceability.
D3	Traffic light thresholds: Green $ Z \leq 2.0$, Amber $2.0 < Z \leq 3.0$, Red $ Z > 3.0$	<code>generate_traffic_light()</code> (line 1329–1416) implements locked thresholds. Worst parameter and direction (delay / reduction / supranormal) reported in message string.	PASS	Confidence values: Green 0.95, Amber 0.85, Red 0.90. Consistent with OSF pre-registration.
D4	Z-score unavailable for contact lens / skin electrodes	<code>ZSCORE_SUPPORTED_ELECTRODES</code> gate at <code>compute_z_scores()</code> (line 1255–1256). Returns empty dict; traffic light returns UNAVAILABLE with <code>NORMATIVE_UNAVAILABLE_TEXT</code> .	PASS	PDF report shows 'No norms' in Z-score column and 'UNAVAILABLE' status for unsupported electrodes.
D5	Paediatric age groups: warning flag and fallback to adult norms	<code>get_age_group_flag()</code> (line 153–164) returns WARNING for 0–12mo, 1–5y; CAUTION for 6–12y. <code>ERGConfig.AGE_GROUPS</code> (line 94–97) includes these strata. However, Baker 2025 normative JSON only contains <code>le35/36to59/ge60</code> strata; paediatric inputs silently fall through to <code>le35</code> .	AMBER	ACTION: PDF report should display a visible paediatric disclaimer block when <code>age_group</code> is a paediatric strata. Currently warning is only written to Layer 4 audit JSON, not to the rendered report.
E — FHIR R4 JSON Output (LOINC 70943-7; Appendix B)				
E1	FHIR Observation resourceType and profile	<code>resourceType = 'Observation'</code> , <code>profile = 'http://hl7.org/fhir/StructureDefinition/Observation'</code> , <code>status = 'final'</code> (line 2370–2377).	PASS	R4 base profile used. Canonical FHIR R4 Observation structure.
E2	LOINC 70943-7 for ERG top-level code	<code>code.coding[0].system = 'http://loinc.org'</code> ; <code>code = '70943-7'</code> ; <code>display = 'Electroretinogram (ERG)'</code> (line 2390–2398).	PASS	Correct LOINC code for electroretinogram as per Appendix B.
E3	Component codes for individual ERG parameters	<code>ERG_PARAMETER_CODES</code> dict uses custom system <code>'erg-api:*</code> ' (line 2302–2315). Comment at line 2299–2301 correctly documents that no validated LOINC codes exist for individual ERG waveform components.	PASS	Custom code system is the correct approach. System URL documented in <code>ERG_CODE_SYSTEM</code> (line 2318).
E4	SNOMED CT interpretation codes for traffic light	<code>SNOMED_INTERPRETATION</code> dict (line 2321–2327): <code>GREEN=17621005</code> (Normal), <code>AMBER=263654007</code> (Borderline), <code>RED=442257004</code> (Abnormal), <code>UNAVAILABLE=410515003</code> (Not available).	PASS	SNOMED codes verified as correct clinical finding codes.
E5	FHIR code system URL is versioned and persistent	<code>ERG_CODE_SYSTEM = 'https://github.com/AI-Fairness-com/ERG-Analysis-API/CodeSystem'</code> (line 2318). This is a GitHub URL without versioning or content negotiation.	AMBER	ACTION: For regulatory submissions, register a versioned <code>CodeSystem</code> resource (e.g. <code>https://ai-fairness.com/fhir/CodeSystem/erg-api 2.4.0</code>). Current URL is

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#	ISCEV Requirement	Pipeline Implementation	Status	Notes / Actions
				acceptable for development but not for clinical deployment.
E6	Patient ID anonymisation	patient_id accepted as string input; no de-identification logic applied within the pipeline. Responsibility delegated to caller (line 2955).	N/A	Correct design — the pipeline is a decision-support tool, not a clinical data system. De-identification is the responsibility of the calling application.
F — PDF Clinical Report (erg_report_generator.py)				
F1	Report title identifies document as decision support only	Title: 'Full-Field Electroretinogram — Clinical Decision Support Report' (erg_report_generator.py line 797). Subtitle cites ISCEV 2022 and Baker et al. (2025) with DOI (line 802–804).	PASS	Title is unambiguous. Subtitle provides normative reference traceability on every page.
F2	Disclaimer: not for standalone clinical use	DISCLAIMER_TEXT (line 658–665) states the report does not constitute a medical diagnosis and must be interpreted by a qualified healthcare professional. Includes specific Green ERG caveat.	PASS	Disclaimer rendered in gold-bordered block on every report page (build_disclaimer_block(), line 668–677).
F3	Normative reference cited on report	Subtitle line: 'Normative reference: Baker et al. (2025), N=407, DOI:10.1007/s10633-025-10009-2' (line 803–804). Full citation also in Layer 4 normative_meta reference_source string.	PASS	Traceability from report to normative source is complete.
F4	Parameter units: amplitudes in µV, times in ms	build_metrics_table() (line 542–627) uses unit 'uV' for amplitude columns and 'ms' for time columns. param_defs list (line 562–568) specifies units per parameter.	PASS	Units consistent with ISCEV convention throughout.
F5	PhNR atypical polarity warning rendered in report	PHNR_ATYPICAL_TEXT (line 680–688) defines warning text. build_phnr_warning_block() renders red-bordered block when phnr_polarity_atypical is True (line 815–817).	PASS	Warning text correctly identifies likely causes (electrode polarity inversion, baseline drift, i-wave contamination).
F6	ISCEV compliance status rendered in quality panel	build_quality_panel() (line 630–655) renders 'Compliant' / 'Non-compliant' based on iscev_compliance.overall_compliant flag from Layer 4 audit (line 752–754).	PASS	ISCEV compliance status visible to clinician on every report.
G — Pipeline Metadata and Version Control				
G 1	PIPELINE_VERSION string consistent throughout	PIPELINE_VERSION = '2.4.0' (line 111). Written to Layer 4 audit JSON and FHIR Observation meta. However, comment header on line 13 reads 'Single source of truth - Version 2.3.2'.	FAIL	ACTION REQUIRED: Line 13 comment must be updated to read 'Version 2.4.0'. This is the only version string inconsistency found. Priority: HIGH — version ambiguity in audit logs is unacceptable for regulatory submissions.
G 2	VERSION_DATE set and written to audit output	VERSION_DATE = '2026-06-18' (line 112). Included in CONFIG.to_dict() (line 166–183).	PASS	Date is plausible for v2.4.0 development timeline.
G 3	Layer 3 Specialist output status clearly declared PENDING	layer_3_specialist status field: 'PENDING — Full implementation requires Chapter 8 (STFT) and Chapter 14 (SHAP)' (line 1523–1531). SHAP Level 2 also declared PENDING (line 2203–2204).	PENDING	This is a known, pre-registered limitation. Correctly labelled; not a compliance failure. Required for Tier 4 / external validation readiness.

5. Remediation Actions

The following actions must be completed before T3-A sign-off. Items are ordered by priority.

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Ref	Priority	Action	Owner	Target
G1	HIGH	Update line 13 comment in <code>erg_v2_4_0.py</code> : change 'Version 2.3.2' to 'Version 2.4.0'.	Hamid Tavakoli	Before T3-A sign-off
A6	MEDIUM	Add runtime flash duration validation: if <code>flash_duration_ms</code> $\geq$ 5.0, set <code>flash_duration_noncompliant</code> = True in Layer 4 audit dict and print warning.	Hamid Tavakoli	T3-A v1.1
D5	MEDIUM	Add a visible paediatric disclaimer block to the PDF report (<code>erg_report_generator.py</code>) when <code>age_group</code> is a paediatric stratum. Mirror the PhNR atypical polarity warning pattern.	Hamid Tavakoli	T3-A v1.1
E5	LOW	Register a versioned FHIR CodeSystem resource for ERG API parameter codes, e.g. https://ai-fairness.com/fhir/CodeSystem/erg-api 2.4.0 . Update <code>ERG_CODE_SYSTEM</code> in <code>erg_v2_4_0.py</code> .	Hamid Tavakoli	Pre-Tier 4

6. Sign-off

This checklist is classified DRAFT. It will be reclassified APPROVED upon:

- Remediation of G1 (FAIL) confirmed by code inspection.
- Hamid Tavakoli review and counter-signature.

Role	Name	Date
T3-A Auditor (H.T)	AI Fairness CIO ERG Pipeline	24 June 2026
Founder & CEO (Approval)	Hamid Tavakoli	26 June 2026

End of T3-A ISCEV Compliance Checklist v1.0 — AI Fairness CIO (Reg. No. 1218464)